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Elevator trench drain

Abstract

A trench drain includes a trench, a grating seat and a grating hingedly connected to the trench. The trench has a base wall with peripheral side walls extending upwardly therefrom and an outwardly-extending peripheral flange. The grating seat is configured to receive the grating. The peripheral side walls of the trench have a pair of opposing elongated slots formed therein and the grating has a pair of posts on opposing ends thereof. Each of the pair of posts on the grating are received in a respective one of the pair of opposing elongated slots of the trench to establish an axis of rotation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation patent application of U.S. patent application Ser. No. 17/228,448, filed on Apr. 12, 2021, which is a continuation of U.S. patent application Ser. No. 16/780,623, filed on Feb. 3, 2020, which is a continuation of U.S. patent application Ser. No. 16/271,324, filed on Feb. 8, 2019, which is a continuation of U.S. patent application Ser. No. 15/689,896 filed on Aug. 29, 2017, which is a continuation of U.S. Ser. No. 15/262,179 filed on Sep. 12, 2016 and which claims the benefit of U.S. Provisional Patent Application No. 62/298,159 filed Feb. 22, 2016, which are hereby incorporated by reference for all purposes as if set forth in their entirety herein.

STATEMENT OF FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not applicable.

TECHNICAL FIELD

(2) This disclosure relates to drains and, in particular, trench drains for use proximate the threshold of an elevator door to prevent water from flowing down an elevator shaft.

BACKGROUND

(3) Nearly all multi-story buildings are required to have fire prevention systems installed. Such fire prevention systems include water lines and sprinklers which, in the event of the detection of a fire, can distribute large volumes of water in the vicinity of a fire. Such fire prevention systems are particularly of importance in multi-story buildings because, in the event of a fire, there may not be an easy way for firefighters at ground level to get several stories up before the fire has had a chance to spread.

(4) However, it is also often the case that multi-story buildings are not well-equipped to accommodate the drainage of such large volumes of water, especially above ground level. In many instances, if the building includes one or more elevators, then the water provided by the fire prevention system collects on the floors above ground level and may have a tendency to flow down the elevator shaft. Because there is the possibility that both the elevator car and other equipment may exist between the floor and the elevator pit, water drainage into the elevator shaft is to be avoided.

(5) Some drainage systems have been developed to address this problem. See e.g., PCT

International Publication No. WO 98/22381 and U.S. Pat. No. 8,800,226. Such systems generally route water laterally into vertical pipes to limit the amount of water entering the open elevator shaft.

SUMMARY

(6) However, the current state of the art drains either need to have great depth to account for the volumes of water that flow therethrough or run the risk of having water overflow from the drain into the elevator shaft.

(7) To prevent draining of water into an elevator and to avoid needing have a deep drain, an improved elevator trench drain is disclosed herein with a low profile that still accommodates high flows of water therethrough. This elevator trench drain may be located at a lower threshold of an elevator door between the floor of the building and the elevator shaft. In this position, any water that collects on the floor (due to, for example, a sprinkler system operating) may be collected in the elevator trench drain and controllably routed to a drainage pipe, rather than simply flowing down into the open elevator shaft. Among other things, this drain can have a low profile, can have a multi-diameter drain opening that builds a head pressure to increase flow out of the drain, and can have a hinged grating for easy installation and access to the trench.

(8) According to one aspect, an elevator trench drain includes a trench and a drain passageway. The trench includes a base wall with peripheral side walls that extend upwardly from the base wall to a trench volume therein. The drain passageway extends downwardly from the base wall of the trench and places the trench volume in fluid communication with a lower opening of the drain passageway (which may be in connection with further drain pipes for routing of the collected water). The drain passageway has a first section proximate the base wall defining a first cross-sectional area and a second section distal the base wall defining a second cross-sectional area. The second cross-sectional area is less than the first cross-sectional area and this reduction from the first cross-sectional area to the second cross-sectional area creates a head pressure that increases the flow out of the lower drain passageway opening during use.

(9) An elevator trench drain of this type may be disposed proximate a lower edge of an elevator door to assist in drainage in the event that water collects on the floor of a multi-story building and to prevent water from draining into the elevator pit in an uncontrolled fashion.

(10) In many forms, the elevator trench drain will have a grating. In some forms, this grating may be hingedly connected to the trench. To provide this hinged connection, the peripheral side walls of the trench may have a pair of opposing elongated slots formed in them and the grating may have a pair of posts on opposing ends thereof. Each of the pair of posts on the grating can be received in a respective one of the pair of opposing elongated slots of the trench to establish an axis of rotation and to permit the assembly/disassembly of the grating to the drain by temporarily axially misaligning the posts relative to the slots. In some forms, the pair of opposing elongated slots may be integrally formed in the peripheral sidewalls of the trench (by welding or casting, for example). It is contemplated that in some forms, each of the slots may positively extend inwardly into the trench from the sidewall. For example, the pair of opposing elongated slots may each be a wall that extends generally perpendicularly from the peripheral side wall of the trench volume and this wall may form a closed loop forming one of the slots into which one of the posts of the grating is received.

(11) With respect to the drain passageway, the first section may have a first distance of downward extension having the first cross-sectional area which is constant along the first distance of downward extension and the second section may have a second distance of downward extension having the second cross-sectional area which is constant along the second distance. In some forms, the first section and the second section may each be tubular and the first and second cross-sectional areas may both be circular. An intermediate section may be between the first section and the second section of the drain passageway in which the intermediate section continuously connects the first section to the second section and reduces in diameter between the two sections. The intermediate

section may extend over a distance of downward extension and gradually tapers inward from the first section to the second section. This intermediate section may be frusto-conical but may have other types of taper. For example, the first section and the second section may have vertically-extending walls and the intermediate section may be angled 45 degrees relative to the vertically extending walls as the intermediate section extends downward and inward from the first section to the second section. Alternatively, it is contemplated that the reduction in cross section could occur in a single step or may have alternative angular tapers.

(12) In some forms, the drain passageway may be centrally aligned in the base wall of the trench while, in other forms, the drain passageway may be offset from the center in the base wall of the trench. Such variations may be made depending on the particular structure in which the drain is used.

(13) In some forms, the peripheral side walls of the trench may include two pairs of linear parallel segments such that the trench is rectangular in shape. In some forms, a flange may be disposed outward of the peripheral side walls. Once installed, floor coverings (for example, tile, carpet, and so forth) may cover this flange. In other forms of the drain, this flange may be absent.

(14) In some forms, to provide sufficient head pressure in the drain passageway, a ratio of a diameter of the first section to a diameter of the second section may be 1.5 or may be more generally in a range of 1.4 to 1.6.

(15) In some forms, a depth of elevator trench drain exclusive of the drain passageway may be 1.75 inches. Among other things, this minimizes the height requirement for the installation of the elevator trench drain into the floor (which takes up the inter-story space between adjacent floors). Despite have relatively shallow depth, the drain passageway of the elevator trench drain may be configured to permit the flow of 100 gallons per minute or more of water therethrough.

(16) These and still other advantages of the invention will be apparent from the detailed description and drawings. What follows is merely a description of some preferred embodiments of the present invention. To assess the full scope of the invention, the claims should be looked to as these preferred embodiments are not intended to be the only embodiments within the scope of the claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a perspective view of an elevator trench drain in which a grating is closed over the trench.

(2) FIG. 2 is a perspective view of the elevator trench drain of FIG. 1 in which a grating is hinged open, revealing the inner volume of the trench and the top side of the drain passageway.

(3) FIG. 3 is a detailed exploded book view of one side the hinge connection between the grating and the trench illustrating how the post of the grating is received in the slot of the trench.

(4) FIG. 4 is a detailed cross-sectional side view of the drain passageway of the elevator trench drain illustrating the reduction in cross-sectional area from the top section to the bottom section of the drain passageway.

(5) FIG. 5 is a detailed top view of drain passageway further illustrating the profile of the upper and lower sections of the drain passageway as seen from above.

DETAILED DESCRIPTION

(6) Referring first to FIGS. 1 and 2, an elevator trench drain **10** is illustrated in closed and opened positions, respectively. An elevator trench drain of this type is designed to be installed along the width of the lower edge or threshold of an elevator door. This drain can provide a drainage pathway for fluids, for example water from sprinkler systems. In the absence of such a drain in a situation in which a sprinkler system has been activated, water from the system might collect on the floor and potentially drain into the elevator or elevator shaft. If that were to happen, the elevator system

could malfunction in the event of an emergency or be otherwise damaged.

(7) The elevator drain **10** is generally rectangular in the exemplary form shown; however, in other forms, the elevator trench drain **10** may have generally different shapes or aspect ratios which may be custom to the particular application. Because the exemplary elevator trench drain **10** is rectangular, it extends between a pair of lateral ends **12** and **14** which define an overall length of the assembly and extends between a pair of forward and backward ends **16** and **18** which define an overall width of the assembly.

(8) In the illustrated assembly, the elevator trench drain **10** has two main parts including a trench **20** and a grating **22**. During use, the water will flow through the grating **22** and collect in the trench **20**, through which the water is subsequently drained or evacuated.

(9) The trench **20** has a base wall **24** with peripheral side walls **26** extending upwardly from the base wall **24**. Because the particular drain **10** is rectangular, the peripheral side walls **26** include a pair of lateral side walls **28** and **30**, a front wall **32**, and a rear wall **34**. In aggregate, the various walls of the trench **20** defining a trench volume **36** inside the trench **20**. In the instant case, this trench volume **36** is generally rectangular in shape being established by the base wall **24**, the pair of lateral side walls **28** and **30**, the front wall **32**, and the rear wall **34**.

(10) The various peripheral side walls **26** of the trench **20** also include an inwardly extending step **38** between an upper peripheral lip **40** and the base wall **24** that defines a grating seat **42** for the reception of the grating **22**. In the form illustrated, this step **38** occurs approximately halfway between an upper surface **44** of the upper peripheral lip **40** and the base wall **24**. During use, water flowing into the trench **20** may also collect (at least to some extent) in the trench volume **36** partially occupied by the grating **22**.

(11) The trench **20** may also include features that assist in the installation of the elevator trench drain **10** into the floor, although such features may not be found in all designs. In the particular form illustrated, the trench **20** includes an outwardly-extending peripheral flange **46** that is offset slightly downward from the upper peripheral lip **40**. This peripheral flange **46** may receive floor coverings (for it, carpet, tile, cast materials such as concrete) over it such that the top of the floor covering is roughly flush with the upper surface **44** of the upper peripheral lip **40**. There may also be anchor straps **48** that extend outwardly past the flange **46** which are employed during the installation of the trench **20** into the surrounding building structure.

(12) With brief forward reference being made to FIG. 3, to hingedly receive the grating **22** in the trench **20**, each of the lateral side walls **28** and **30** may include a slot **50** formed therein such that the slots **50** collectively define a pair of opposing slots. In the particular form illustrated in FIG. 3, the slot **50** is formed above the step **38** in the lateral side walls **28** and **30** proximate the rear wall **34**. To accommodate entry of the respective post **52** on the grating **22**, these slots **50** are elongated in the front to rear direction such that the grating **22** may be slightly twisted relative the trench **20** and the eventual hinged axis of rotation to provide sufficient clearances for assembly of the posts **52** into the slots **50**. In the form illustrated, the slots **50** are integrally formed in the peripheral side walls **26** of the trench **20** and are formed by a wall **54** that extends generally perpendicularly from the peripheral side wall **26** of the trench **20** and forms a closed loop that establishes the slot **50**.

(13) With particular reference now being made to FIG. 4, the trench **20** includes a drain passageway **56** that extends downwardly the base wall **24** of the trench **20**. As illustrated, the drain passageway **56** is centrally located along the width and length of the trench **20**. However, it is contemplated that the drain passageway **56** might be located otherwise such as, for example, at one end of the base wall **24** of the trench **20** or offset from center in one or both of the length and width directions. This drain passageway **56** places the trench volume **36** (at an upper opening **58** formed in the base wall **24**) in fluid communication with a lower opening **60** of the drain passageway **56**. The lower opening is typically connected in use to a drain pipe such as a 4 inch drain pipe for further routing of the drained water.

(14) As best seen in FIGS. 4 and 5, the drain passageway **56** has multiple sections as it extends

downwardly from the upper opening **58** to the lower opening **60**. In the particular form illustrated, there are three sections including an upper section **62** proximate the upper opening **58** in the base wall **24**, a lower section **64** proximate the lower opening **60** of the drain passageway **56**, and an intermediate section **66** between the upper section **62** and the lower section **64**. The upper section **62** and the lower section **64** each have a constant respective cross-sectional area over their axial height or distance of extension. In the illustrated embodiment, the upper section **62** is circular having a diameter of 6 inches and the lower section **64** is circular having a diameter of 4 inches (to match standard drain pipe which is connected to the outlet). The intermediate section **66** between the upper section **62** and the lower section **64** includes an approximately 45 degree taper that transitions the diameter between the respective sections **62** and **64** above and below it. Although the taper is illustrated as extending entirely from the first section to the second section, it is contemplated that there may be more than one differently angled tapered regions or radially extending step in this intermediate section. While circular cross-sections are found in the illustrated design as can be best seen in FIG. 5, it is contemplated that other drain cross sections might also be used if those reduce in cross-section area from the upper section to the lower section.

(15) Among other things, the profile of this drain passageway **56** helps to improve the flow rate of water collecting in the trench **50** through the drain passageway **56**. The upper section **62** creates a head pressure which causes the acceleration of the water flowing downward into the lower section **64**. Although it is contemplated that the intermediate section **66** might be omitted (i.e., that there could simply be a flat step between sections **62** and **64** which is perpendicular to the central axis of the drain passageway **62**), the taper or angling of the intermediate section **66** can further enhance the flow from the upper section **62** to the lower section **64** because it assists in directing the flow of the water in a non-turbulent manner from the upper section **62** into the lower section **64**. Among other things, the reduction in diameter to improve and increase flow rate can help enable a low profile (i.e., thin) design so that the elevator trench drain **10** may be installed in smaller spaces without increasing the depth of the floor to accommodate for the drain.

(16) Some exemplary and non-limiting dimensions are now provided for the trench **20**. The height of the trench **20** is contemplated as being typically 1.75 inches overall (exclusive of the drain passageway **56**), although other depths may also be used depending on the installation context or needs of the customer. A typical width may be 13.5 inches overall with an 11.4375 inch grating seat area. The finished perimeter flange may be approximately 1 inch from the upper peripheral lip **40**. The overall length of channel may typically be in a range of 38 to 120 inches. The drain passageway **56** is illustrated as being hydraulically engineered from 6 inches to 4 inches in a funnel style outlet to increase head pressure on the 4 inch section to promote flow (with flow in excess of 100 gallons per minute being targeted in many conditions to meet code and achieved using this passageway structure). This ratio of the diameters from the first section to the second section is approximately 1.5 (6 inches to 4 inches), but it is contemplated that other similar ratios may also be used, for example, in the range of 1.3 to 1.7 or more narrowly in the range of 1.4 to 1.6. The entire trench construction may be provided in Type 304 stainless steel but may also be in Type 316 stainless steel in certain applications or based on customer preference.

(17) The grating **22** is also generally rectangular in shape to match and fit into the grating seat **42** of the trench **20**. The exemplary grating **22** includes a first plurality of lengthwise-extending bars **68** which are joined to a second plurality of widthwise-extending supports **70** to form a grid. However, in other forms, the grating pattern may be different or otherwise embellished to provide a desired aesthetic appearance. On the rear side of the grating **22**, there are a pair of oppositely facing posts **52** that are located for reception into the slots **50** of the trench **20** as illustrated in FIG. 3. FIGS. 3 and 4 also illustrate that each of the first plurality of lengthwise-extending bars **68** includes a bottom surface **100**, facing and positioned a first distance **102** from the base wall **24**, a top surface **104**, opposite the bottom surface **100** and spaced a second distance **106** from the base wall **24** greater than the first distance, and a pair of side surfaces **108** extending between the bottom surface

100 and the top surface **104**. As shown in FIG. 3, the side surfaces **108** are sub-divided into two portions **112**, **116**. The lower portion **112** of the side surface **108** is oriented substantially parallel to the front or first wall **32** and the upper portion **116** of the side surface **108** is oriented oblique to the front wall **32**. More specifically, FIG. 3 shows that the upper portion **116** of the side surfaces **108** are oriented oblique to the front wall **32** in a vertical orientation (e.g., measured vertically). FIG. 3 also illustrates that each of the lengthwise extending bars **68** include a lower edge **120** where the bottom surface **100** meets the side surface **108** and an upper edge **124** where the top surface **104** meets the side surface **108**. In the illustrated embodiment, the lower edge **120** is spaced a second distance **106** from the base wall **24** while the upper edge **124** is spaced a first distance **102** from the base wall **24** that is greater than the second distance **106**. FIGS. 1 and 3 also show that the upper edge **124** is both parallel to and positioned closer to the front wall **32** than the lower edge **120**. The lower edge **120** is also parallel the front wall **32**. Each bar **68** also defines a longitudinal axis **132** therethrough that is oriented parallel to the front wall **32**.

(18) As noted above, by angularly twisting the axis of the posts **52** (which are co-axial with one another) they can be slid into the slots **50** of the trench **20** to provide a hinged connection to open the grate as generally illustrated in FIG. 2. There may be some range of motion restrictions depending on the depth of the grating seat **42** and the placement of the posts **52** on the grating **22**.

(19) The grating **22** may be a stainless steel wire grating and may be approximately 2 inches shorter than the specified channel length to accommodate installation of the grating **22** into the trench **20** and provide some side clearances. As with the trench **20**, the grating **22** may be manufactured from Type 304 stainless steel or Type 316 stainless steel. In one exemplary form, the grating **22** may features an open area of 64.9 square inches per linear foot of grating. Wires, bars, and supports may be held by press fit and welded to support trusses (i.e., the bars **68** may be press fit and welded into supports **70** or vice versa). Some gratings may be a fabricated stainless steel slotted grate with an open area of 35.9 square inches per linear foot. In some forms, grating height may be 29/32 inches tall and grating width may be 11.375 inches wide.

(20) Accordingly, an elevator trench drain is disclosed that accommodates high flow rates (up to and exceeding 100 gallons per minute) without having a deep trench depth or necessarily having multiple outlets to accommodate slower flows. The improved drain passageway permits high flow rates without compromising other dimensions of the product or complicating the drain system with additional, multiple fluid connections.

(21) It should be appreciated that various other modifications and variations to the preferred embodiments can be made within the spirit and scope of the invention. Therefore, the invention should not be limited to the described embodiments. To ascertain the full scope of the invention, the following claims should be referenced.

Claims

1. A trench drain comprising: a trench including a base wall, a first wall extending from the base wall, and a second wall extending from the base wall, the base wall, the first wall, and the second wall at least partially defining a trench volume therein, the trench having a trench height, a trench length, and a trench width, the trench length is greater than the trench width, and the trench length is at least 60 times the trench height; a drain passageway extending through the base wall and including a distal end opposite the base wall, the distal end having an outlet, the outlet having an inner diameter that is no greater than 4 inches; and a grate at least partially positionable within the trench volume, the grate configured to allow fluid to pass therethrough.

2. The trench drain of claim 1, wherein the grate includes a plurality of bars configured to allow fluid to pass therebetween, wherein at least one of the plurality of bars includes a bottom edge positioned a first distance from the base wall and a top edge positioned a second distance from the base wall greater than the first distance, and wherein the top edge is positioned closer to the first

wall than the bottom edge.

3. The trench drain of claim 1, wherein the drain passageway includes a first section connected to the base wall defining a first diameter and a second section distal the base wall defining a second diameter that is less than the first diameter.

4. The trench drain of claim 3, wherein a ratio of the first diameter to the second diameter is between 1.3 and 1.7.

5. The trench drain of claim 3, wherein a ratio of the first diameter to the second diameter is between 1.4 and 1.6.

6. The trench drain of claim 3, wherein a reduction from the first diameter to the second diameter creates a head pressure that increases flow out of the outlet.

7. The trench drain of claim 1, wherein the trench height is 1.75 inches.

8. The trench drain of claim 1, wherein the drain passageway is configured to permit flow of at least 75 gallons per minute of water therethrough.

9. A trench drain comprising: an elongated trench defining a trench volume therein, the elongated trench having a trench height, a trench length, and a trench width, the trench length is at least eight times the trench width, and the trench length is at least 60 times the trench height; a drain passageway open to the trench volume, the drain passageway extending away from the trench volume and including a distal end spaced apart from the trench volume, the distal end having an outlet, the outlet having an interior diameter that is no greater than 4 inches; and a grate at least partially positioned within the trench volume, the grate including a plurality of elongated openings through which fluid can enter the trench volume, the elongated openings having an opening length and an opening width, the opening length being greater than the opening width, and the trench length oriented substantially parallel to the opening length.

10. The trench drain of claim 9, wherein the drain passageway is configured to permit flow of at least 75 gallons per minute of water therethrough.

11. The trench drain of claim 9, wherein the drain passageway includes a first section connected to the trench volume defining a first diameter and a second section distal the trench volume defining a second diameter that is less than the first diameter.

12. The trench drain of claim 11, wherein a ratio of the first diameter to the second diameter is between 1.3 and 1.7.

13. The trench drain of claim 11, wherein a ratio of the first diameter to the second diameter is between 1.4 and 1.6.

14. The trench drain of claim 11, wherein a reduction from the first diameter to the second diameter creates a head pressure that increases flow out of the outlet.

15. A trench drain comprising: an elongated trench defining a trench volume therein, the elongated trench having a trench height, a trench length, and a trench width, the trench length being greater than the trench width, the trench height is no greater than 1.75 inches, and the trench length is at least 60 times the trench height; a drain passageway open to the trench volume, the drain passageway extending away from the trench volume and including a distal end spaced apart from the trench volume; and a grate at least partially positioned within the trench volume, the grate including a plurality of elongated openings through which fluid can enter the trench volume, the elongated openings having an opening length and an opening width, the opening length being greater than the opening width, and the opening length being oriented substantially parallel to the trench length.

16. The trench drain of claim 15, wherein the drain passage way includes an upper section defining a first diameter proximate the trench volume and a lower section defining a second diameter proximate the distal end, and wherein the second diameter is less than first diameter.

17. The trench drain of claim 16, wherein a ratio of the first diameter to the second diameter is between 1.3 and 1.7, and wherein the distal end has an inner diameter that is no greater than 4 inches.

18. The trench drain of claim 16, wherein a reduction from the first diameter to the second diameter creates a head pressure that increases flow out of the distal end.

19. The trench drain of claim 15, further comprising: a base wall having peripheral side walls extending upwardly therefrom and an outwardly-extending peripheral flange that together define at least a portion of a perimeter of the trench; and a grate seat disposed between an upper peripheral lip and the base wall, and wherein the grate is hingedly connected to the trench, wherein the grate seat is configured to receive the grate and the peripheral side walls of the trench have a pair of opposing elongated slots formed therein and the grate has a pair of posts on opposed ends thereof, wherein each of the pair of posts on the grate are received in a respective one of the pair of opposing elongated slots of the trench to establish an axis of rotation, and wherein the axis of rotation is movable relative to the trench.

20. The trench drain of claim 19, wherein the elongated slots are sized so that the posts can be inserted into and removed from the slots by moving the axis of rotation relative to the trench.
